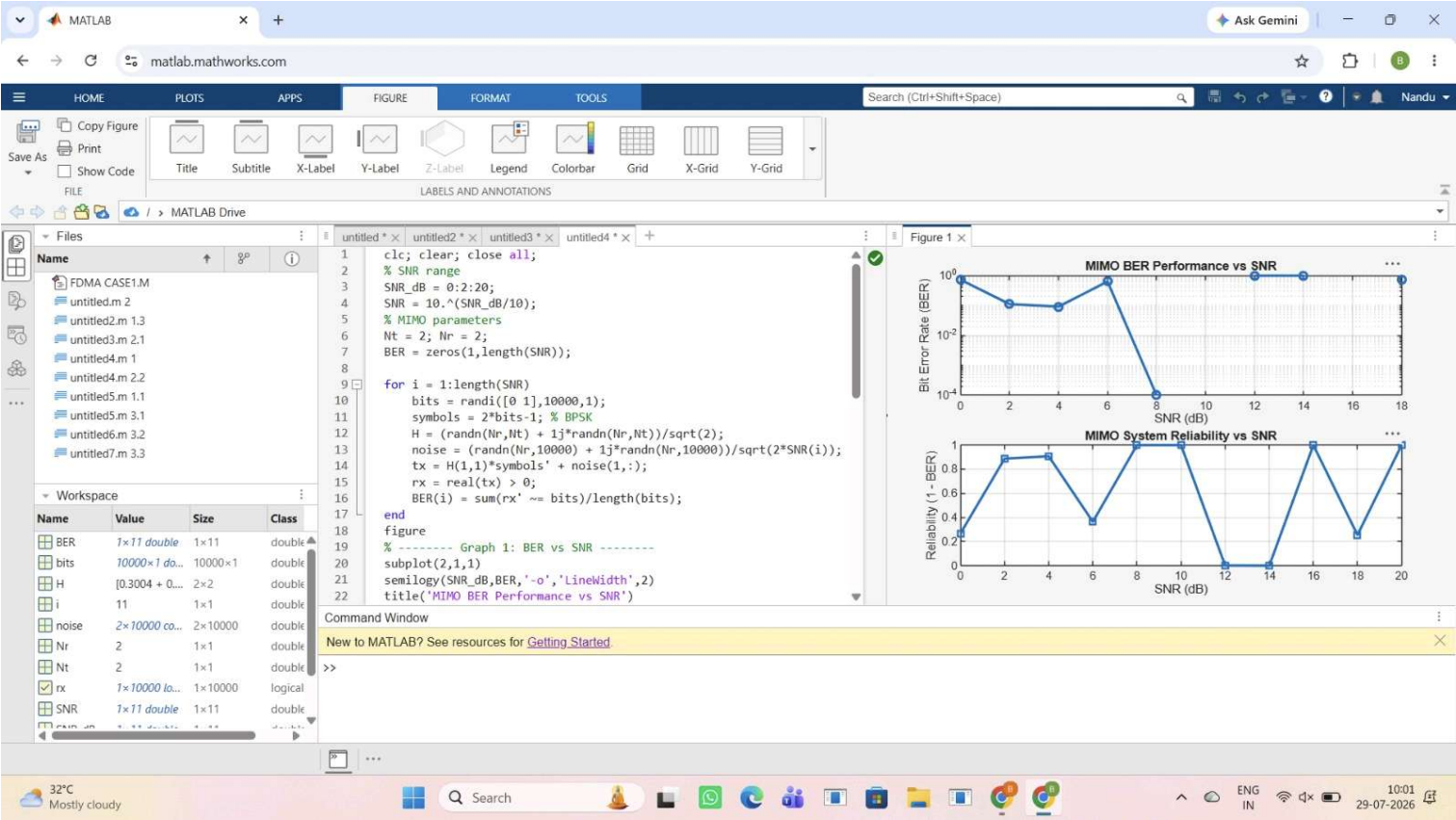
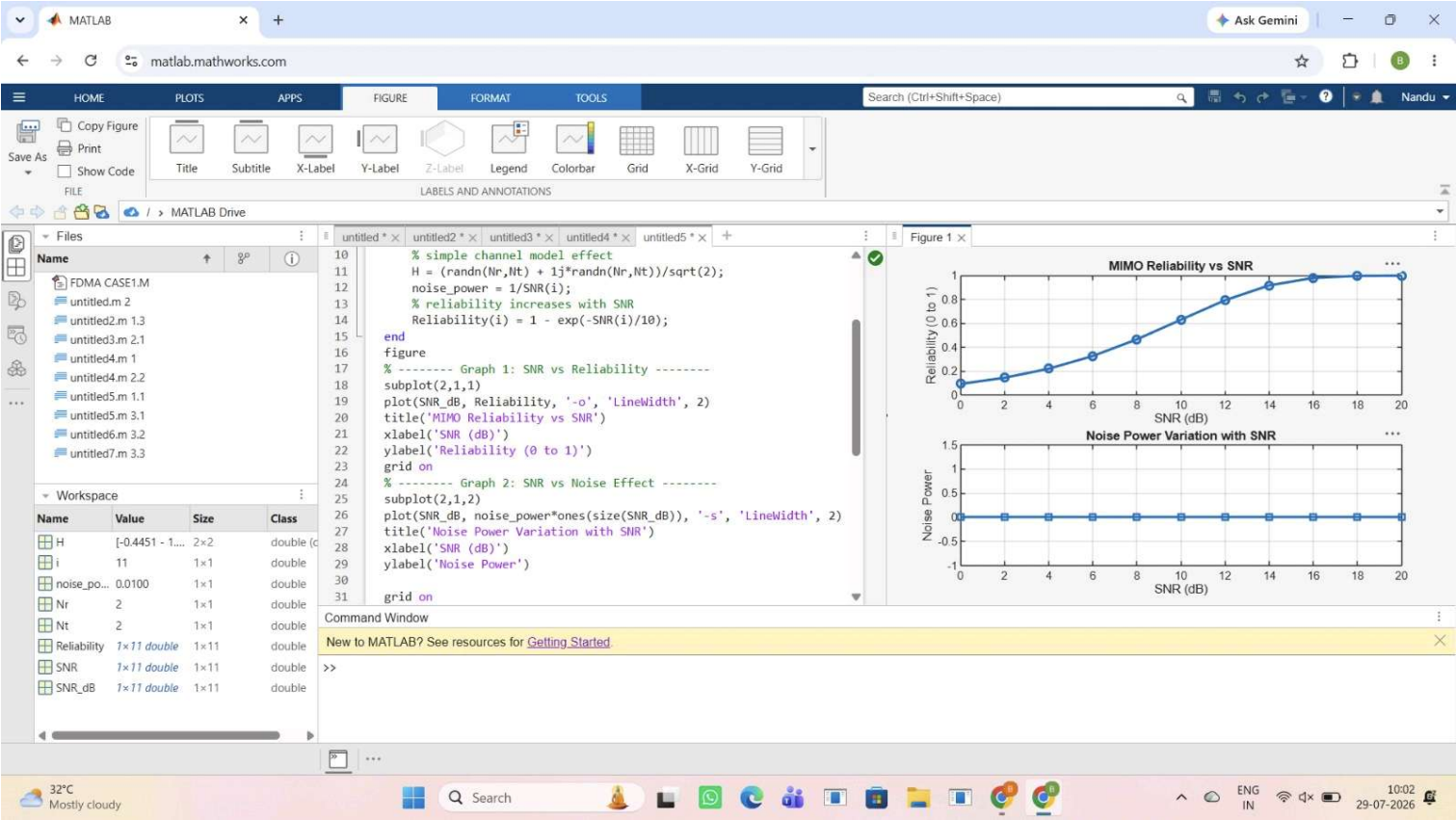






MATLAB

matlab.mathworks.com

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LABELS AND ANNOTATIONS

Files

FDMA CASE1.M

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untitled4.m 2.2

untitled5.m 1.1

untitled5.m 3.1

untitled6.m 3.2

untitled7.m 3.3

Workspace

Name	Value	Size	Class
alloc_RB	[10,12,8,15,5]	1x5	double
mod_eff	[2,4,6,4,2]	1x5	double
rb_bw	180000	1x1	double
through...	[3.6000,8.64...	1x5	double
users	[1,2,3,4,5]	1x5	double

```

9 mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
10 % Throughput calculation (Mbps)
11 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
12
13 % Display results
14 disp('User Throughput (Mbps):')
15 disp(throughput)
16 figure
17 % ----- Graph 1: Throughput per User -----
18 subplot(2,1,1)
19 plot(users, throughput, '-o','LineWidth',2)
20 title('OFDMA User Throughput')
21 xlabel('User Index')
22 ylabel('Throughput (Mbps)')
23 grid on
24 % ----- Graph 2: RB vs Throughput -----
25 subplot(2,1,2)
26 plot(alloc_RB, throughput, '-s','LineWidth',2)
27 title('Resource Blocks vs Throughput')
28 xlabel('Allocated Resource Blocks')
29 ylabel('Throughput (Mbps)')
30 grid on

```

Figure 1

The figure consists of two subplots. The top subplot, titled 'OFDMA User Throughput', shows the throughput (Mbps) for five users. The x-axis is 'User Index' (1 to 5) and the y-axis is 'Throughput (Mbps)' (0 to 10). The data points are approximately (1, 4.0), (2, 8.64), (3, 8.64), (4, 10.08), and (5, 4.0). The bottom subplot, titled 'Resource Blocks vs Throughput', shows the throughput (Mbps) versus the number of allocated resource blocks. The x-axis is 'Allocated Resource Blocks' (5 to 15) and the y-axis is 'Throughput (Mbps)' (0 to 10). The data points are approximately (10, 4.0), (12, 8.64), (8, 8.64), (15, 10.08), and (15, 4.0).

Command Window

New to MATLAB? See resources for [Getting Started](#).

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